MATHEMATICS 4052 · PAPER 2 · PRACTICE PAPER IN THE GCE FORMAT

9 questions · 90 marks · 2 hours 15 minutes · AI-authored draft (seed 1) — not a past-year paper · Answer ALL questions in the spaces provided.

Name: _____ Date: _____

1 **(a)** Solve the inequality $5 - 3(x - 2) > 2x + 1$. [2]

(b) Simplify $\left(9p^{4}q^{-2}\right)^{\frac{1}{2}} \times 2p^{-3}q^{5}$, giving your answer in positive index form. [2]

(c) Make t the subject of the formula $v = \sqrt{\frac{k-3t}{t}}$. [3]

(d) Solve the equation $\frac{4}{x-1} + \frac{3}{x+2} = 2$. Give your solutions correct to two decimal places. [4]

2 Mei runs a small café.

(a) Mei buys a coffee machine from Italy for €2 450. The exchange rate is €1 = S$1.48.
Goods and Services Tax (GST) of 9% is charged on the price in Singapore dollars.
Calculate the total amount, in Singapore dollars, that Mei pays for the machine. [2]

(b) Mei increased the selling price of a cup of latte by 8%. The new selling price is $5.94.
Calculate the selling price of a cup of latte before the increase. [2]

(c) The cost of making a cup of latte is made up of the cost of coffee beans, the cost of milk and other costs in the ratio 5 : 4 : 3. The coffee beans for one cup cost $1.25.
Calculate the percentage profit Mei makes on a cup of latte sold at $5.94. [2]

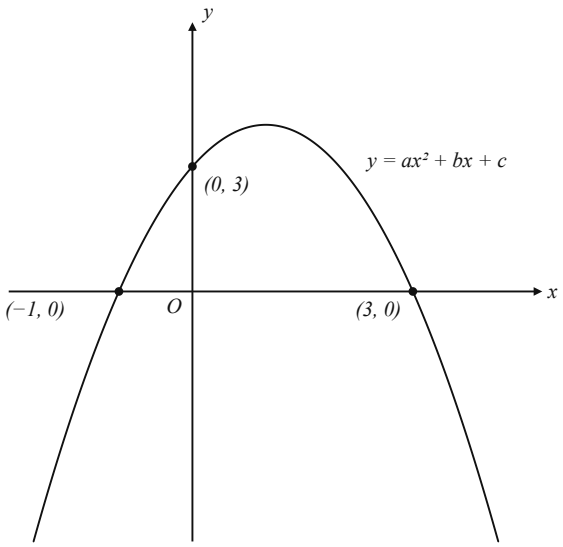
(d) Mei spends \$180 on Grade A coffee beans, which cost \$$x$ per kilogram.
Grade B coffee beans cost \$5 per kilogram more than Grade A coffee beans.
For \$180, Mei would get 3 kg less of Grade B coffee beans than of Grade A coffee beans.

(i) Form an equation in x and show that it simplifies to $x^2 + 5x - 300 = 0$. [3]

(ii) Solve the equation $x^2 + 5x - 300 = 0$. [1]

(iii) Hence find the cost of 1 kilogram of Grade B coffee beans. [1]

3 The diagram shows a sketch of the curve $y = ax^2 + bx + c$.
The curve crosses the x-axis at the points $(-1, 0)$ and $(3, 0)$ and crosses the y-axis at the point $(0, 3)$.



(a) Find the value of a, of b and of c. [3]

(b)

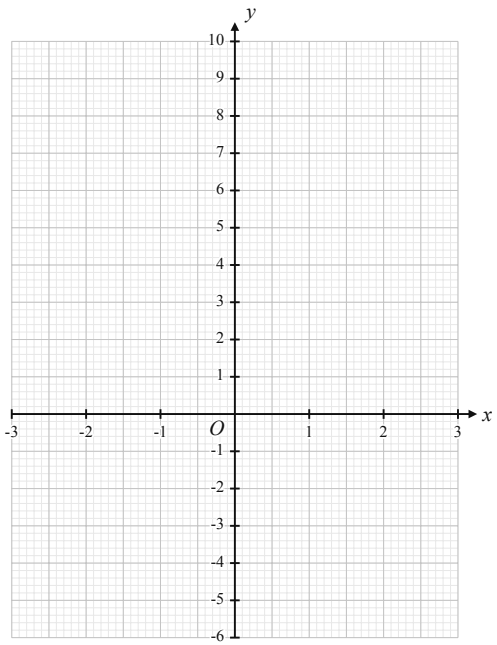
(i) Write the equation of the curve in the form $y = k - (x - h)^2$, where h and k are integers. [2]

(ii) Hence write down the coordinates of the maximum point of the curve. [1]

(c) The point P lies on the curve and has x-coordinate 4.
Find the equation of the straight line that passes through P and the maximum point of the curve. [3]

4 The table shows some values of x and the corresponding values of y for $y = x^2 + \frac{3}{x}$.

x	-3	-2.5	-2	-1.5	-1	-0.5	0.5	1	1.5	2	2.5	3
y	8	5.05	2.5		-2	-5.75	6.25	4	4.25	5.5		10



(a) Complete the table of values. [2]

(b) On the grid, draw the graph of $y = x^2 + \frac{3}{x}$ for $-3 \leq x \leq 3$. [3]

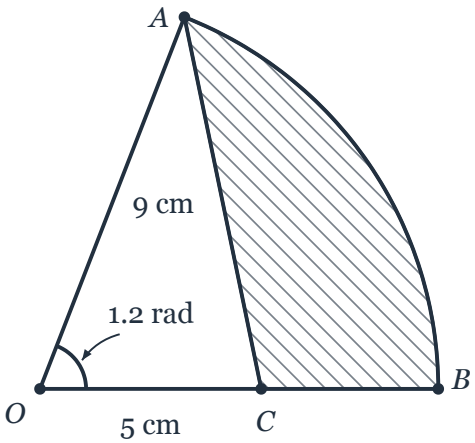
(c)

(i) By drawing a tangent, use your graph to find the gradient of the curve at the point where $x = 2$. [2]

(ii) Use your graph to explain why the equation $x^2 + \frac{3}{x} = 3$ has only one solution. [1]

(iii) By drawing a suitable straight line on the grid, solve the equation $x^3 - 2x^2 - 3x + 3 = 0$. [3]

5 The diagram shows a sector OAB of a circle, centre O and radius 9 cm.
Angle $AOB = 1.2$ radians.
C is the point on OB such that $OC = 5$ cm.
The region bounded by AC, CB and the arc AB is shaded.



(a) Calculate AC. [2]

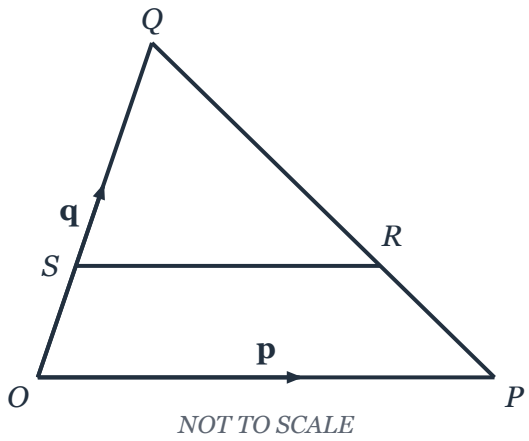
(b) Calculate angle OAC.
Give your answer in degrees. [2]

(c) Calculate

(i) the perimeter of the shaded region, [2]

(ii) the area of the shaded region. [3]

6 The points A, B and C have coordinates $(-4, -1)$, $(2, 7)$ and $(10, -1)$ respectively.



NOT TO SCALE

(a) Find the equation of the line which passes through C and is parallel to AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers. [3]

(b) Calculate the length of AB. [2]

(c) Calculate the area of triangle ABC. [2]

(d) OPQ is a triangle.
$\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OQ} = \mathbf{q}$.
R is the point on PQ such that $PR : RQ = 1 : 2$.
S is the point on OQ such that $OS : SQ = 1 : 2$.

(i) Express $\overrightarrow{OR}$ in terms of $\mathbf{p}$ and $\mathbf{q}$, as simply as possible. [2]

(ii) Express $\overrightarrow{SR}$ in terms of $\mathbf{p}$ and $\mathbf{q}$, as simply as possible. [1]

(iii) Hence find $\dfrac{\text{area of triangle } QSR}{\text{area of triangle } QOP}$. [1]

7 The table shows the distribution of the distances, d km, cycled by the members of a cycling club in one week.

Distance (d km)	$0 < d \leqslant 20$	$20 < d \leqslant 40$	$40 < d \leqslant 60$	$60 < d \leqslant 80$	$80 < d \leqslant 100$
Frequency	5	x	18	13	7

The estimated mean distance cycled is 50 km.

(a) Find the value of x. [2]

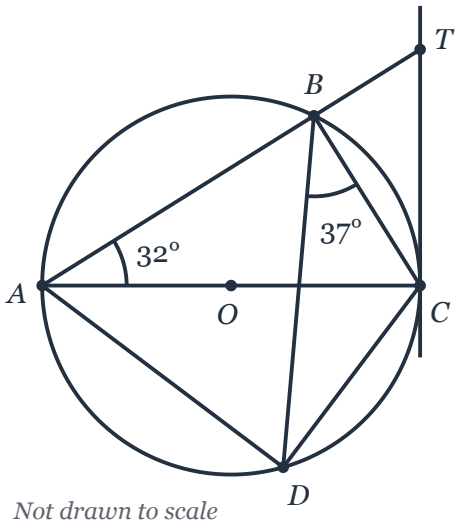
(b) Calculate an estimate of the standard deviation of the distances. [2]

(c)

(i) One of the members is chosen at random. Find the probability that this member cycled more than 60 km. [1]

(ii) Two of the members are chosen at random. Find the probability that exactly one of them cycled more than 60 km. Give your answer as a fraction in its lowest terms. [3]

8 A, B, C and D are points on a circle, centre O.
AOC is a diameter of the circle.
The tangent to the circle at C meets AB produced at T.
Angle $BAC = 32°$ and angle $CBD = 37°$.



Not drawn to scale

(a) Find, giving a reason for each step of your working,

(i) angle ACB, [2]

(ii) angle BCT, [2]

(iii) angle CAD. [1]

(b) Explain why AD is not parallel to BC. [1]

(c) The radius of the circle is 6 cm.

(i) Show that triangle ABC is similar to triangle CBT.
Give a reason for each statement you make. [2]

(ii) Calculate BT. [2]

9 Mr Tan is deciding whether to install a rainwater tank to water the plants in his garden.
He finds the following information.

Rainwater tank
The tank is a cylinder with internal diameter 80 cm and internal height 120 cm.
Price, including installation: \$420
Expected lifespan: 10 years

Collecting rainwater
Rain falling on the roof of the garden shed drains into the tank.
The roof is a rectangle 3.6 m by 2.5 m.
80% of the rain that falls on the roof reaches the tank.
Average rainfall: 175 mm per month

Watering the garden
Mr Tan waters his garden with a hose for 12 minutes every day.
The hose delivers 9 litres of water per minute.

Monthly water tariff

First 40 m^3	\$2.74 per m^3
Each m^3 above 40 m^3	\$3.69 per m^3

1 m^3 = 1000 litres

(a) Calculate the capacity of the tank.
Give your answer in litres. [2]

(b) In a month with 30 days, calculate

(i) the volume of water, in litres, Mr Tan uses to water his garden, [1]

(ii) the volume of rainwater, in litres, that reaches the tank. [2]

(c) Before installing the tank, Mr Tan's household uses 43 m^3 of tap water each month.
Assume that all the rainwater that reaches the tank is used to water the garden instead of tap water.
Is it worth Mr Tan installing the rainwater tank?
Justify your answer with calculations. [5]

END OF PAPER

ANSWER KEY

1 (a) $x < 2$

(b) $\frac{6q^4}{p}$

(c) $t = \frac{k}{v^2+3}$

(d) $x = 3.71$ or $x = -1.21$

2 (a) S$3952.34

(b) $5.50

(c) 98%

(d)(i) $\frac{180}{x} - \frac{180}{x+5} = 3$, which simplifies to $x^2 + 5x - 300 = 0$ (shown)

(d)(ii) $x = 15$ or $x = -20$

(d)(iii) $20 (taking $x = 15$; $x = -20$ rejected as a price cannot be negative)

3 (a) $a = -1$, $b = 2$, $c = 3$

(b)(i) $4 - (x - 1)^2$

(b)(ii) $(1, 4)$

(c) $y = -3x + 7$

4 (a) When $x = -1.5$, $y = 0.25$; when $x = 2.5$, $y = 7.45$

(b) Correct smooth curve in two branches: the left branch through $(-3, 8)$, $(-2.5, 5.05)$, $(-2, 2.5)$, $(-1.5, 0.25)$, $(-1, -2)$, $(-0.5, -5.75)$ and the right branch through $(0.5, 6.25)$, $(1, 4)$, $(1.5, 4.25)$, $(2, 5.5)$, $(2.5, 7.45)$, $(3, 10)$, with no curve drawn between $x = -0.5$ and $x = 0.5$; the right branch has a minimum near $(1.1, 3.9)$

(c)(i) 3.25 (accept 2.9 to 3.6)

(c)(ii) The line $y = 3$ cuts the curve at one point only (where $x \approx -2.1$): the right-hand branch of the curve lies entirely above $y = 3$, its lowest point being at $y \approx 3.9$, so there is no second intersection.

(c)(iii) Line $y = 2x + 3$; $x = -1.45$, $x = 0.75$, $x = 2.7$ (accept $-1.55 \leq x \leq -1.4$, $0.7 \leq x \leq 0.85$, $2.6 \leq x \leq 2.8$)

5 (a) $AC = 8.57$ cm

(b) Angle $OAC = 33.0°$

(c)(i) 23.4 cm

(c)(ii) 27.6 cm^2

6 (a) $4x - 3y = 43$

(b) $AB = 10$ units

(c) 56 units^2

(d)(i) $\overrightarrow{OR} = \frac{2}{3}\mathbf{p} + \frac{1}{3}\mathbf{q}$

(d)(ii) $\overrightarrow{SR} = \frac{2}{3}\mathbf{p}$

(d)(iii) $\frac{4}{9}$

7 (a) $x = 17$

(b) 22.8 km

(c)(i) $\frac{1}{3}$

(c)(ii) $\frac{80}{177}$

8 (a)(i) $58°$

(a)(ii) $32°$

(a)(iii) $37°$

(b) If AD were parallel to BC, angle CAD would equal angle ACB (alternate angles); but $37° \neq 58°$, so AD is not parallel to BC.

(c)(i) Angle ABC = angle $CBT = 90°$ (angle in a semicircle; ABT is a straight line), angle BAC = angle $BCT = 32°$; two pairs of equal angles, so the triangles are similar (AA).

(c)(ii) 3.97 cm

9 (a) 603 litres

(b)(i) 3240 litres

(b)(ii) 1260 litres

(c) Rainwater collected 1260 litres $= 1.26$ m^3 per month (less than the 3240 litres used, so all of it replaces tap water). Household use falls from 43 m^3 to 41.74 m^3, still above 40 m^3, so the whole saving is at the higher rate: saving $= 1.26 \times \$3.69 = \$4.65$ per month ($\backslash 4.6494).Overthe10-yearlifespan$:4.6494 \times 120 = \557.93 (accept \$558.00 from $\$4.65 \times 120$), which exceeds the \$420 price, so YES it is worth installing (net saving about \$138; equivalently the tank pays for itself in $420 \div 4.6494 = 90.3$, i.e. 91 months, about 7.5 years, within its 10-year lifespan). Also accept NO with the same figures and a valid reason (a net saving of only about \$138 over 10 years, about \$14 a year, is small; rainfall varies and the tank may overflow or run dry). A candidate who prices the saving at \$2.74 per m^3 (\$3.45 per month, \$414.29 over 10 years, giving NO) has used the wrong tier and does not earn the tier or conclusion marks.